Hanka Goralija

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EDUCATION

École Polytechnique Fédérale de Lausanne **2024 - present (expected July 2026)**
- Master's in Robotics *(focus on Machine Learning and Optimization)*
- GPA 5.42/6.0
- Thesis supervisor: prof. Volkan Cevher

University of Sarajevo - Faculty of Electrical Engineering Sarajevo **2021 - 2024**
- BSc in Automatic Control and Electronics
- GPA 9.97/10.0

EXPERIENCE

EPFL **September 2025 - January 2026**
- Research Assistant @LASA
  - Contributing to the IEEE-funded RAS-U initiative by developing structured robotics course material and coordinating content with international contributors.
  - Working on Sampling-Based Motion Planning and Graph Theory.

Maxon **July 2025 - September 2025**
- Software Engineer Intern
  - Developed a Python-based testing framework with TwinCAT integration, automating motor control experiments and migrating the company's testing bench to the industry-standard TwinCAT platform.

PROJECTS

- Evolution Strategies for Deep RL pretraining @Swiss Data Science Center Summer semester 2025
  - Studied Evolution Strategies (ES) and implemented ES-based pretraining pipelines for DRL tasks (Breakout, Flappy Bird, MuJoCo).
  - Evaluated the impact of ES pretraining vs random initialization on DRL performance.

- Semester project at Risk Analytics and Optimization Chair at @EPFL Summer semester 2025
  - Extended the Federated Conditional Stochastic Gradient algorithm (FCSG) using Multilevel Monte Carlo estimators.
  - Applied the method to invariant logistic regression and mean-squared Bellman error problems, analyzing variance reduction and convergence accuracy.

- Semester project at Biorobotics Lab at @EPFL Winter semester 2024
  - Developed ROS/Gazebo and IsaacGym pipelines for a quadruped performing object-throwing while balancing on rear legs.
  - Designed trajectory interpolation for throwing motion and trained PPO-based RL models for control, comparing simulation results across frameworks.

TEACHING EXPERIENCE

Olympiads in Informatics in Bosnia and Herzegovina **August 2021 – present**
- Mentored national teams preparing for international informatics competitions, teaching and emphasizing algorithmic thinking and problem-solving skills.
- Leader of BIH team at international olympiads: EJOI 2023 Georgia, EGOI 2024 Netherlands.

Faculty of Electrical Engineering Sarajevo **October 2022 – February 2024**
- Undergraduate teaching assistant for the courses "Introduction to programming" and "Programming techniques".

LANGUAGES

Bosnian (native), English (fluent, TOEFL iBT 101), German (B1)

AWARDS AND COMPETITIONS

- Robotics Master Admission Award at EPFL 2024
- Top 100 in IEEEXtreme 16.0 programming competition 2022
- Bronze medal in EGOI 2021 (European Girls' Olympiad in Informatics) 2021
- First place in Federal Math competition 2021
- Bronze medal in IGO 2019 (Iranian Geometry Olympiad) 2019
- Participation in International Olympiad in Informatics as a member of national team 2019
- Bronze medal in Junior Balkan Olympiad in Informatics 2017
- First place in National Junior Informatics Competition 2017

EXTRACURRICULAR ACTIVITIES

- Summer and Winter Math Camps in Bosnia and Herzegovina @Mathematical Society of Canton Sarajevo and Croatia @Croatian Mathematical Society and MNM 2017 - 2021
- Mathematics School for Gifted Students @Mathematical Society of Canton Sarajevo 2017 - 2021
- Maritime Robotics In Blue Economy (MARBLE) Training Course - Trieste, Italy May 2023

SKILLS & INTERESTS

- **Skills:** Problem solving, Competitive programming, Python, C++, PyTorch, MATLAB, ROS, Isaac Gym
- **Interests**: Machine Learning, Optimization, Reinforcement Learning, Robotics